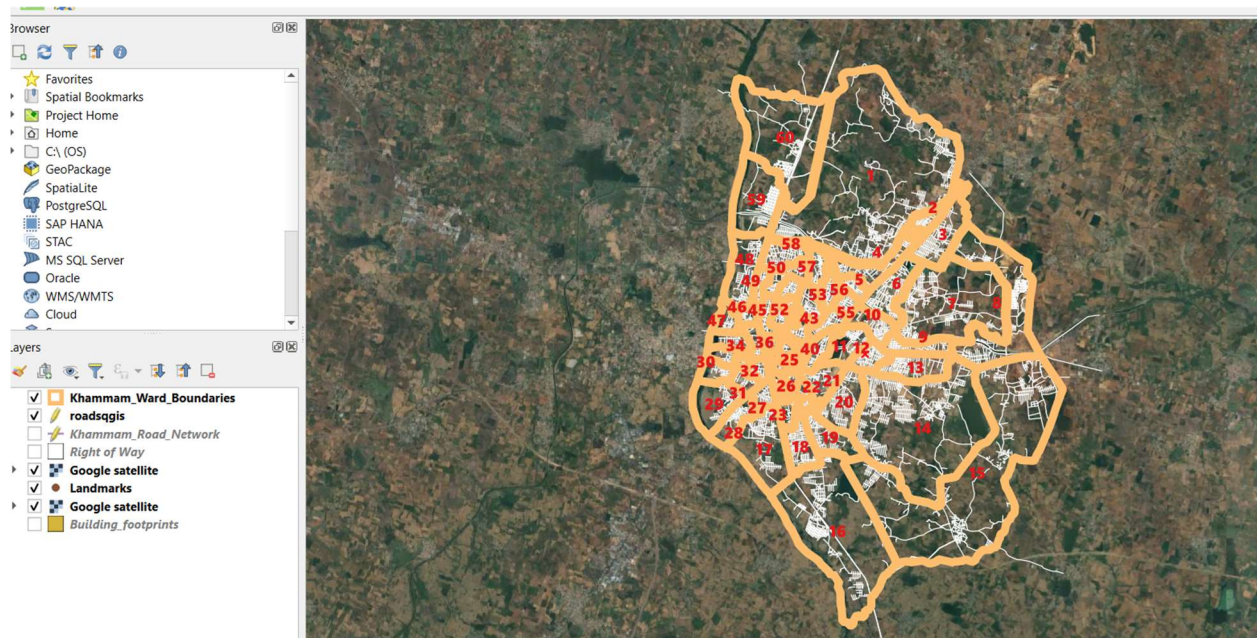
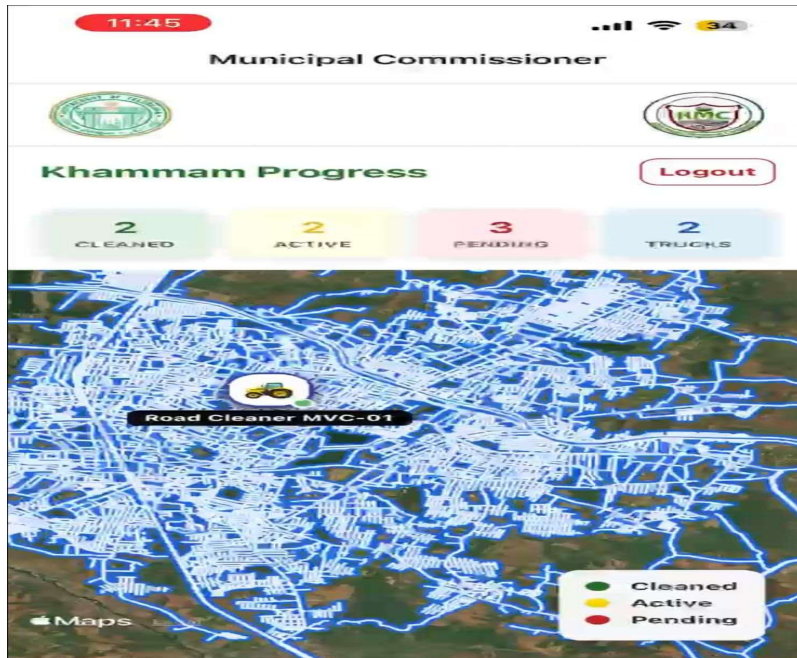




Smart GIS-Based Road and Drainage Monitoring System

The proposed project is a **Smart GIS-Based Road and Drainage Cleaning Monitoring System** designed for municipal corporations to monitor sanitation activities in real time. The system uses **QGIS, PostgreSQL/PostGIS, mobile applications, and GIS dashboards** to improve transparency, accountability, and operational efficiency in city cleaning management.

In this system, the complete road and drainage network of the city is mapped using GIS data obtained from municipal shapefiles or OpenStreetMap. The road network is divided into smaller segments, and each segment is assigned to a sanitation worker (Jawan). Every segment contains attributes such as worker name, cleaning status, timestamp, geo-tagged photos, and supervisor review.

The system follows a three-stage workflow:

- **Red** indicates roads pending for cleaning.
- **Yellow** indicates roads cleaned by workers after uploading geo-tagged photo proof.
- **Green** indicates roads verified and approved by supervisors.

Field workers use a mobile application to upload before/after cleaning photographs along with GPS coordinates and timestamps. The uploaded data is stored in a **PostgreSQL + PostGIS spatial database** through a **Flask backend API**. Supervisors receive notifications, verify the uploaded proof, add remarks, and approve the work through the dashboard.

The GIS dashboard displays ward boundaries, road segments, drainage systems, worker assignments, and live cleaning status on satellite maps. Municipal officers and collectors can monitor ward-wise progress, identify pending roads, and generate reports for better decision-making.

The project aims to create a transparent and technology-driven sanitation management system for smart cities. It reduces manual monitoring, improves accountability of workers, enables real-time supervision, and supports efficient urban governance through spatial analytics and GIS technology.